

Compound Interest – Solutions

LET – Mathematics and Statistics

1. (a) **Interest is not compounded (simple interest)**

Solution: Using the simple interest formula $A = P(1 + rt)$:

$$A = 5000(1 + 0.04 \times 5)$$

$$A = 6000$$

Answer: £6000.00

(b) **Interest is compounded yearly**

Solution: Using the formula $A = P(1 + r/n)^{nt}$ with $P = 5000, r = 0.04, t = 5, n = 1$:

$$A = 5000 \left(1 + \frac{0.04}{1}\right)^{1 \times 5}$$

$$A = 5000(1.04)^5$$

$$A \approx 6083.26$$

Answer: £6083.26

(c) **Interest is compounded monthly**

Solution: Using the formula with $P = 5000, r = 0.04, t = 5, n = 12$:

$$A = 5000 \left(1 + \frac{0.04}{12}\right)^{60}$$

$$A \approx 6104.98$$

Answer: £6104.98

(d) **Interest is compounded continuously**

Solution: Using the formula $A = Pe^{rt}$ with $P = 5000, r =$

$0.04, t = 5$:

$$A = 5000 \times e^{0.2}$$

$$A \approx 6107.01$$

Answer: £6107.01

2. (a) **The depreciation is compounded quarterly**

Solution: Using the formula with $P = 250, r = -0.12, t = 8, n = 4$:

$$A = 250 \left(1 + \frac{-0.12}{4} \right)^{4 \times 8}$$

$$A = 250(0.97)^{32}$$

$$A \approx 94$$

Answer: £94

(b) **The depreciation is compounded continuously**

Solution: Using the formula $A = Pe^{rt}$ with $P = 250, r = -0.12, t = 8$:

$$A = 250 \times e^{-0.96}$$

$$A \approx 96$$

Answer: £96

3. *Solution:* Let's calculate the final amount for a principal (P) of £1000 over 10 years ($t = 10$) for each option.

I. 8% compounded yearly ($r = 0.08, n = 1$)

$$A = 1000(1.08)^{10} \approx 2158.92$$

II. 7.9% compounded monthly ($r = 0.079, n = 12$)

$$A = 1000 \left(1 + \frac{0.079}{12}\right)^{120} \approx 2197.70$$

III. 7.8% compounded continuously ($r = 0.078$)

$$A = 1000e^{0.078 \times 10} \approx 2181.47$$

Comparing the three final amounts, the second option yields the highest return.

Answer: Option II (7.9% compounded monthly) gives the greatest return (£2197.70).

4. *Solution:* We need to find the rate r where the outcomes are equal. Let P be the principal.

Annual Compounding = Continuous Compounding

$$P(1.12)^{20} = Pe^{20r}$$

The principal P cancels from both sides.

$$(1.12)^{20} = e^{20r}$$

To solve for r , we take the natural logarithm ($\ln$) of both sides.

$$\ln(1.12^{20}) = \ln(e^{20r})$$

$$20 \ln(1.12) = 20r$$

$$r = \ln(1.12)$$

$$r \approx 0.113328...$$

Answer: 11.4%